

Excel for Data Analytics

Session Flow: Introduction | Formatting | Aggregation | Conditionals | Validation

Part 1: Excel Introduction

Excel is a powerful spreadsheet tool used for data storage, analysis, and visualization. In the field of Data Analytics, it is the go-to tool for data cleaning, quick statistical analysis, and creating business dashboards.

Excel Structure

Component	Description	Example
Workbook	The entire Excel file	Sales_Report.xlsx
Worksheet	An individual sheet inside the workbook	January_Data
Cell	The intersection of a row and a column	B4

Column	Vertical markers identified by letters	A, B, C...
Row	Horizontal markers identified by numbers	1, 2, 3...

Part 2: Basic Data Formatting

Formatting ensures that data is professional and easy to interpret at a glance.

1. **Number Formatting:** Convert raw numbers into Currency (₹), Percentages (%), or Dates.
 2. **Font & Cell Styling:** Use **Bold**, *Italics*, or Cell Fill Colors to highlight top-performing sales or headers.
 3. **Alignment & Structure:** Use **Merge & Center** for titles and **All Borders** to define your data tables clearly.
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Part 3: Aggregation Functions

Aggregation functions allow you to summarize large datasets into single, meaningful values.

Function	Purpose	Formula Example
SUM	Adds all values in a range	=SUM(B2:B10)
AVERAGE	Calculates the arithmetic mean	=AVERAGE(B2:B10)
COUNT	Counts cells containing numbers	=COUNT(B2:B10)
COUNTA	Counts all non-empty cells (text + numbers)	=COUNTA(A2:A10)
MAX	Finds the highest value	=MAX(B2:B10)

MIN	Finds the lowest value	=MIN(B2:B10)
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Part 4: Conditional Functions

Conditional logic allows Excel to make "decisions" based on data values.

The IF Function

Structure: =IF(condition, value_if_true, value_if_false)

- **Pass/Fail Example:** =IF(B2>=50, "Pass", "Fail")
 - **Sales Target Example:** =IF(A2>=10000, "Target Achieved", "Target Missed")
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Part 5: Data Validation

Data Validation is a critical step in **Data Cleaning**. it prevents users from entering inconsistent data (e.g., entering "Male", "M", and "male" in the same column).

Key Features:

- **Dropdown Lists:** Restrict entries to a specific list (e.g., "Male, Female").
- **Number Restrictions:** Ensure marks are only entered between 0 and 100.
- **Error Alerts:** Show a custom message when a user enters invalid data.

Practice Exercise

Dataset:

| Name | Marks |

| :--- | :--- |

| A | 75 |

| B | 40 |

| C | 60 |

| D | 90 |

Tasks:

1. **Total:** Use **SUM** to find total marks.
 2. **Average:** Use **AVERAGE** to find the mean.
 3. **High/Low:** Use **MAX** and **MIN**.
 4. **Logic:** Create a "Result" column using **=IF(B2>=50, "Pass", "Fail")**.
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Homework Assignment

Dataset:

| Product | Sales |

| :--- | :--- |

| A | 500 |

| B | 300 |

| C | 700 |

| D | 200 |

Tasks:

1. Calculate Total and Average Sales.
2. Identify the Highest and Lowest sales figures.
3. **Advanced:** Create a "Performance" column where Sales > 400 is "High Sales" and everything else is "Low Sales".